



CAREERFORGE: AN AI-BASED CAREER PATH RECOMMENDATION SYSTEM FOR ENGINEERING STUDENTS

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Abstract : Deciding career path is primary concern for graduating engineers. The career path expert system named as CareerForge represents a comprehensive solution to the critical problem of career confusion among engineering students. So, as a helping hand, we propose an expert system that presents a comprehensive solution to the critical problem of career confusion among just graduated engineers. By leveraging artificial intelligence, data analytics, and personalized learning approaches, the system transforms the career exploration journey from overwhelming and uncertain to structured and achievable. The platform's unique combination of branch-specific mapping, intelligent recommendation algorithms, actionable learning roadmaps, multilingual AI assistance, and community support creates an ecosystem that empowers student aspirants to make informed career decisions confidently. By addressing the needs of student aspirants from all engineering branches, including those uncomfortable with coding and those from core engineering disciplines, the system ensures inclusive career guidance. The implementation of this expert system has the potential to significantly improve employment outcomes for engineering graduates, enhance industry-academia alignment, and contribute to building a more skilled and satisfied technical workforce. By providing clear pathways, realistic expectations, and continuous support, the Career Path Expert System bridges the gap between engineering education and successful career outcomes, ultimately fulfilling the promise of technical education in shaping prosperous professional futures.

Keywords: Career guidance, engineering students, artificial intelligence, machine learning, personalized learning, career assessment, skill development, job readiness, recommendation system, educational technology, career counselling, multilingual support, data analytics.

I. INTRODUCTION

In the modern era of rapid technological advancement and digital transformation, the scope of engineering careers has expanded significantly across multiple domains such as software development, data science, artificial intelligence, core engineering, management and entrepreneurship. While this diversity created numerous opportunities, it also introduces a major challenge for engineering graduate students-selecting the most suitable career path. Many student aspirants experience confusion due to a lack of proper guidance, limited awareness of available options, and uncertainty about their own strengths, interests and long-term goals[1]. Traditional career counselling methods are often insufficient to address these challenges, as they rely heavily on manual evaluation, limited interaction, and generalized advice. Moreover, these methods may not keep pace with rapidly changing industry demands and emerging job roles. As a result, student aspirants may make uninformed decisions, leading to skill mismatches, reduced employability, and dissatisfaction in their professional life[2].

To overcome these limitations, there is a growing need for intelligent, scalable, and personalized Career Path Expert System for Engineering graduates. This system leverages advanced technologies such as artificial intelligence machine learning, and data analytics to provide customized career recommendations. By analysing individual student profiles- including academic performance, skills, interests, and preferences- the system generates tailored suggestions that align with both personal capabilities and industry requirements.

In addition to career recommendations, the system offers structured learning roadmaps that guide student aspirants step-by-step in achieving their desired career goals. These roadmaps include relevant courses, certifications, projects and internship opportunities, thereby enhancing students' practical knowledge and job readiness. The inclusion of multilingual support further ensures accessibility and usability for students from diverse linguistic backgrounds, making the system more inclusive and effective.[3]

Another key aspect of the proposed system is its ability to bridge the gap between academia and industry. By incorporating real-time insights into market trends and skill demands, the platform enables student aspirants to make informed decisions and stay competitive in the job market. It also supports student aspirants from all engineering disciplines, including those from non-IT backgrounds or those who may not be inclined toward coding, ensuring equal opportunities for career growth.[4]

In conclusion, this research aims to develop an intelligent and user-centric career guidance system that simplifies the decision-making process, enhances employability, and contributes to the development of a skilled and future-ready engineering workforce.

II. LITERATURE REVIEW

Career guidance systems have evolved significantly with the advancement of technology. Traditional counselling methods often rely on manual assessment and limited interaction, which may not effectively address the diverse needs of engineering students. Several studies have explored the use of intelligent systems to enhance career decision-making. Research on AI-based recommendation systems highlights their ability to analyse user data and provide personalized suggestions. These systems utilize machine learning algorithms to evaluate students' skills, interests, and academic performance. According to recent studies, recommendation systems improve decision accuracy and user satisfaction compared to conventional methods. Additionally, career assessment platforms have incorporated data analytics to identify industry trends and skill demands. This enables students to align their learning paths with market requirements. However, many existing systems lack comprehensive support for all engineering branches and often focus primarily on IT-related careers. Furthermore, multilingual support and user-friendly interfaces are identified as critical factors for accessibility. Studies show that students are more likely to engage with systems that provide guidance in their preferred language and offer clear, structured roadmaps. Despite these advancements, there remains a gap in developing a unified platform that integrates personalized recommendations, branch-specific guidance, and continuous learning support. The proposed system aims to address these limitations by offering a holistic and inclusive career guidance solution.

III. METHODOLOGY

The proposed CareerForge system follows a structured and data-driven methodology to provide personalized career recommendations for engineering students. The system integrates artificial intelligence and machine learning techniques to analyze user data and generate accurate and relevant career guidance.

A. Data Collection

The first step involves collecting relevant data from students through forms or questionnaires. This data includes:

- Academic performance (CGPA, subjects)
- Technical and non-technical skills
- Extracurricular activities
- Interests and career preferences

This information forms the foundation for generating personalized recommendations, as highlighted in recent AI-based career guidance systems [5], [6].

B. Data Preprocessing

The collected data is cleaned and pre-processed to remove inconsistencies, missing values, and irrelevant information. The preprocessing phase includes:

- Data normalization
- Feature selection
- Categorization of student profiles

Proper preprocessing ensures improved accuracy of machine learning models [6].

C. Feature Extraction and Analysis

In this phase, key features such as skills, interests, and academic strengths are extracted from the dataset. These features are analyzed to understand student profiles and identify patterns that influence career decisions.

Machine learning models rely on these extracted features to perform accurate classification and recommendation tasks [7].

D. Recommendation and Career Mapping Module

The core component of the system is the Recommendation and Career Mapping Module, which utilizes machine learning algorithms to suggest suitable career paths for students.

The system analyzes and compares student profiles—including academic performance, skills, and interests—with predefined career datasets to recommend the most appropriate career options. In addition to recommendations, the module maps different engineering branches to a variety of career opportunities.

It provides detailed insights such as:

- Suitable job roles (e.g., Software Developer, Data Analyst, Core Engineer)
- Required technical and soft skills for each career path
- Current industry trends and job market demand

This integrated approach helps students clearly understand the relationship between their academic background and potential career opportunities, enabling them to make informed career decisions.[7]

E. Learning Roadmap Generation

Based on the recommended career path, the system generates a structured roadmap that includes:

- Online courses and certifications
- Suggested projects
- Internship opportunities
- Skill development strategies

Such personalized learning pathways improve student readiness and career alignment [4].

F. User Interface and Interaction

The system provides an interactive and user-friendly interface where students can:

- Input their data
- View career recommendations
- Access learning roadmaps

Additionally, multilingual support and chatbot assistance enhance usability and accessibility for diverse users [5].

G. Evaluation and Performance Analysis

The system is evaluated based on:

- Accuracy of recommendations
- User satisfaction
- System efficiency

Feedback from users is collected to improve system performance. Continuous updates ensure the system remains aligned with evolving industry requirements [7].

IV. SYSTEM ARCHITECTURE

The proposed CareerForge system is designed using a modular architecture to ensure efficient data flow, scalability, and accurate career recommendations. The system consists of the following key modules:



Figure 1. System Architecture

1. User Input Module

This module collects essential user data such as academic performance, skills, interests, and career preferences through forms or questionnaires. Accurate data collection is critical for generating personalized recommendations in AI-based career guidance systems [5], [6].

2.Data Processing (Preprocessing) Module

The collected data is pre-processed to remove inconsistencies and prepare it for analysis. This includes data cleaning, normalization, and feature selection. Effective preprocessing significantly improves the performance and accuracy of machine learning models [6].

3. Recommendation Engine and Career Mapping Module

This is the core component of the system where machine learning algorithms analyze processed data and generate suitable career recommendations. Techniques such as classification algorithms (Decision Trees, Naïve Bayes), collaborative filtering, and content-based filtering are used to match user profiles with relevant career options [7], [8].

This module maps engineering disciplines to various career paths such as software development, data science, core engineering roles, and higher education. It helps bridge the gap between academic qualifications and industry requirements [7].

4. Skill Gap Analysis Module

This module compares the user's existing skills with the required skills for recommended careers and identifies missing competencies. It provides suggestions for skill improvement, enabling users to better prepare for their chosen career paths [8].

5. Learning Resource Module

This module provides curated learning materials such as online courses, tutorials, certifications, and practice platforms. It helps users follow structured learning paths aligned with their career goals [5].

6. Database Module

The database stores user profiles, career datasets, skill requirements, and learning resources. It ensures efficient data retrieval and supports real-time recommendation generation. Regular updates to the database enhance system relevance and adaptability [7].

7. Feedback and Improvement Module

This module collects user feedback on recommendations and system usability. The feedback is used to improve the recommendation model, enhance accuracy, and personalize future suggestions [8].

8. Authentication and Security Module

This module ensures secure user authentication and protects sensitive data. It includes login systems, access control, and data encryption mechanisms to maintain privacy and security [6].

9. User Interface Module

The user interface provides an interactive platform for users to input data, view recommendations, and track progress. A user-friendly interface enhances accessibility, engagement, and overall system usability [5].

V. Results and Discussion

The proposed CareerForge system was evaluated using sample student data, and the results indicate that the AI-based recommendation engine effectively provides relevant career suggestions aligned with user profiles. Similar AI-driven systems have demonstrated improved accuracy in matching user preferences with suitable career paths [5], [7].

Students reported increased clarity in career decision-making and a better understanding of the skills required for their chosen paths. This aligns with findings that personalized recommendation systems enhance student engagement and decision-making capabilities [8].

The system also demonstrated scalability and adaptability, making it suitable for deployment across a large number of users. The integration of real-time data and intelligent algorithms further improves the relevance of recommendations [7].

However, the effectiveness of the system depends heavily on the accuracy and completeness of input data. Additionally, regular updates are necessary to keep the system aligned with evolving industry trends and job market requirements, as emphasized in recent studies on AI-based educational systems [5], [6].

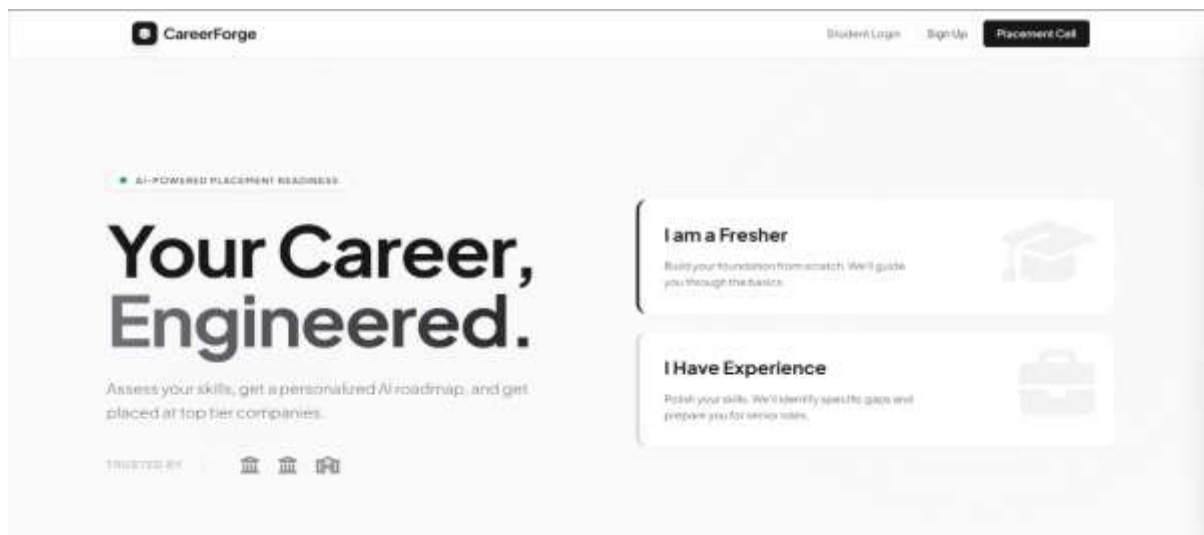


Figure 2. Landing Page

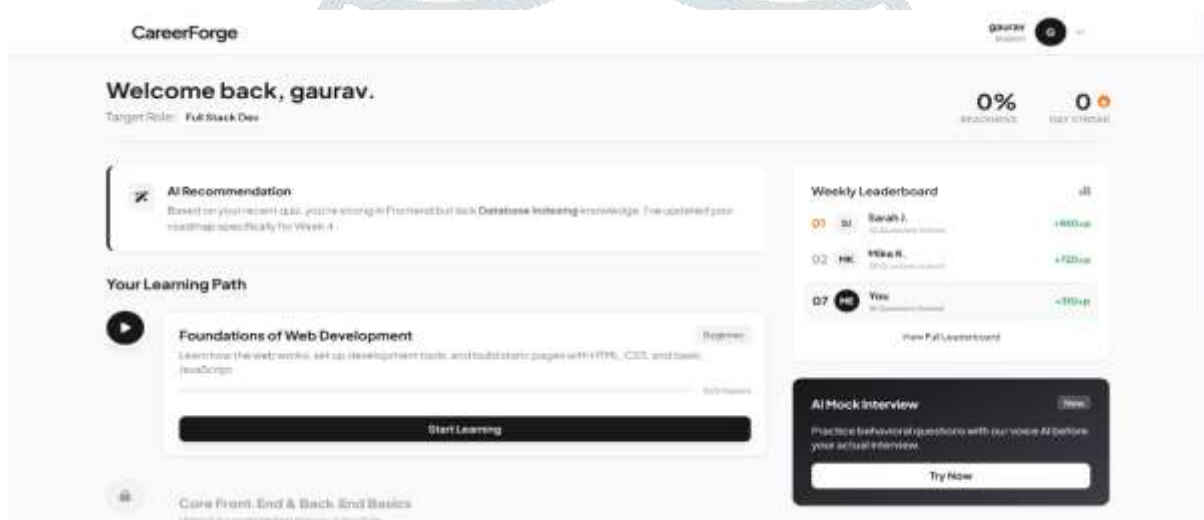


Figure 3. Home Page

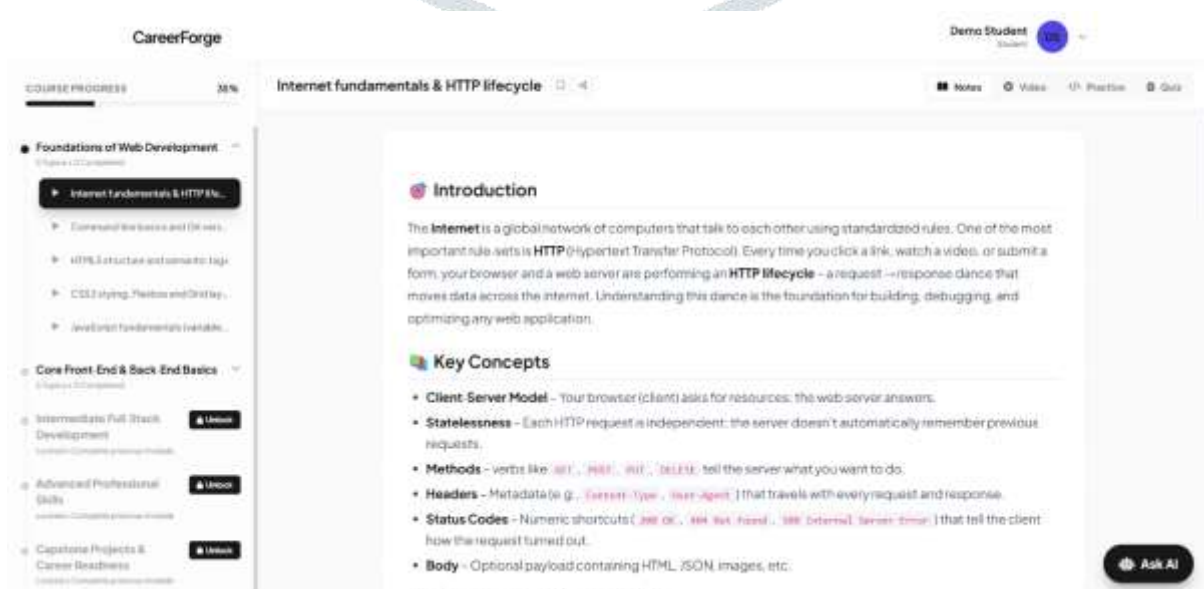


Figure 4. Learning Dashboard

6. Conclusion

The proposed CareerForge system provides an effective solution to the problem of career confusion among engineering students by leveraging artificial intelligence and machine learning techniques. The system successfully delivers personalized career recommendations based on individual skills, interests, and academic performance, which is consistent with recent advancements in AI-based recommendation systems [5], [7].

By integrating structured learning roadmaps and career mapping, the system enhances students' understanding of required skills and improves their decision-making abilities. Studies have shown that such intelligent systems significantly improve student engagement and career planning efficiency [8], [5].

Moreover, the system bridges the gap between academic learning and industry requirements by incorporating data-driven insights and real-time trends, thereby improving employability outcomes [7]. Its scalable and user-friendly design makes it suitable for widespread adoption across different engineering disciplines.

However, the effectiveness of the system depends on the quality of input data and continuous updates to align with evolving job market demands, as highlighted in recent research [6]. Future enhancements may include real-time job integration, advanced predictive analytics, and industry collaboration to further improve system accuracy and usability.

“The proposed system also contributes to multiple Sustainable Development Goals (SDGs), including SDG 4 (Quality Education) by enhancing personalized learning, SDG 8 (Decent Work and Economic Growth) by improving employability, SDG 9 (Industry, Innovation and Infrastructure) through AI-driven innovation, and SDG 10 (Reduced Inequalities) by ensuring inclusive career guidance for students from diverse backgrounds.”

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